

Joseph Mateo B Borbor  
8-Rosal

Initial Draft for Project Proposal

# “Safe Measures, Safe Labwork”



To be submitted to: Ma'am Erica Vinas

Sep 26, 2026

## COMPUTER SCIENCE 2

### Coding in a Connected World: Development of Computational Thinking Skills

#### QUARTER 1

---

## Parts of the Project Proposal

### I. Project Title

“Safe Measures, Safe Labwork”

### II. Problem Statement

When scholars are doing research in the lab facilities, it is necessary to follow precautions and warnings. Sometimes, students become careless and do unnecessary actions that may endanger fellow scholars and themselves. To ensure that the students follow and remember what precautions to follow, this can be their guide. It can be tweaked and adjusted based on the labwork. My project aims to help remind students of basic safety precautions and prepare the lab equipment before doing research.

### III. Project Objectives

- To develop a program that will identify possible safety hazards and needed equipment for the lab based on the users inputs.
- To provide safety reminders to students.

### IV. Planned Features

Check whether a lab activity will involve hazards such as:

Heat

Open Flame

Chemicals

Glassware

Electricity

Identify the safety risks

Recommend appropriate (PPE)

Give safety reminders to students

### V. Planned Inputs and Outputs

What will the user do or give (inputs)? What will the system show, do, or give back (outputs)?

Inputs:

Hazards Present (Yes/No)

Outputs:

Warnings, Safety precautions, Overall Safety Status, Total amount of hazards.

## COMPUTER SCIENCE 2

### Coding in a Connected World: Development of Computational Thinking Skills

#### QUARTER 1

---

VI. Logic Plan:

Pseudocode:

Print "Laboratory Risk Safety Checker"

Set risklevel = 0

Chemicals = Input "Are chemicals being used? (yes/no)"

Heat = "Will heat be used? (yes/no)"

Flame = "Will an open flame be used? (yes/no)"

Glass = "Will glassware be used? (yes/no)"

Electricity = "Will electricity be used? (yes/no)"

Print "" (For blank space)

if Chemicals = "yes"

then

Print "WARNING: Chemical hazard detected."

Print "Wear safety goggles and appropriate protective equipment."

risklevel + 1

if Heat = "yes"

then

Print "WARNING: Heat hazard detected."

Print "Use heat compatible equipment and handle hot materials carefully."

risklevel + 1

if Flame= "yes"

then

Print "WARNING: Open Flame detected."

Print "Keep flammable materials away from the flame. "

risklevel + 1

## COMPUTER SCIENCE 2

### Coding in a Connected World: Development of Computational Thinking Skills

#### QUARTER 1

---

```
if Glass= "yes"
then
Print "WARNING: Glassware is being used."
Print "Handle glassware carefully and check for cracks."
risklevel + 1
```

```
if Electricity= "yes"
then
Print "WARNING: Electric materials are being used."
Print "Keep liquids away from electrical equipment."
risklevel + 1
```

```
Print "" (For blank space)
```

```
If risklevel = 0
then
Print "No hazards detected."
Print "Always follow your teacher's laboratory instructions."
```

```
Else If risklevel <= 2
then
Print "Safety Status: Low Risk"
Print "Review the safety precautions before starting."
```

```
Else
Print "Safety Status: HIGH RISK"
Print "Review all safety precautions and ask your teacher before
proceeding. "
```

```
Print "" (For blank space)
```

```
Print "Total Hazards Detected: ", risklevel
```